

Close Encounters of the Unidentified Kind: Investigation into UFO Sightings Across North America

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Abstract

This template is provided for papers at the bachelor's and master's programs in Linguistics at the University of Bergen, Norway. It demonstrates the use of $\text{\LaTeX}$, a document preparation system which is quite suited to scientific papers and theses. This document is not intended to give a realistic example of linguistic content, but attempts to demonstrate many different textual elements that are common in a term paper or a bachelor's thesis.

1 Introduction

The UFO phenomenon has been a topic of interest and debate for decades, with numerous reports and sightings recorded around the world. In recent years, advancements in technology and the widespread availability of online reporting systems have made it easier than ever to collect and analyze data on UFO sightings. The National UFO Research Center (NUFORC) has compiled a comprehensive database of descriptive data on UFO sightings in North America from 1969 to 2019, which is available in both CSV and JSON formats. This report aims to uncover new insights into this mysterious phenomenon and provide a more comprehensive understanding of the UFO phenomenon to the scientific community and the general public. The specific areas of this data we'll be exploring are explained below.

The first question which came to mind when confronted with our data was how the number of sightings varied geographically. In essence, we severely doubted that the number of sightings across the country was spread evenly geographically. As such, we theorized that there may be a correlation between the number of sightings and areas with high air-traffic, airports, military bases, and other aerial-relevant bases. A previous study conducted in 2018 by the Mutual UFO Network (MUFON) found a positive correlation between the two, suggesting that military activity in these areas may be related to an increased likelihood of UFO sightings. (Blundell, 2018).

Similarly, we hypothesized that the number of sightings every year was likely to increase over time. This topic was addressed whilst simultaneously considering the impact of technology, as we suspected technology to be a confounding factor in this scenario. A study by Pilkington (2017) found a significant increase in UFO sightings after the widespread availability of smartphones, suggesting that technology does play a role in the reporting of UFO sightings.

Yet another frequency-centric problem we explored was the variance in sightings among the four seasons of the year. This was something we weren't entirely sure would reveal any sort of variation, however the trend was actually quite interesting when interacting with the variable of time.

2 Data Analysis

The dataset used for this study was obtained from the National UFO Reporting Center (NUFORC), an organization dedicated to the collection and documentation of UFO sightings in North America. With over 49 years of operation, NUFORC has amassed a database of more than 170,000 sightings. The data analyzed in this study spans from 1969 to 2019 and includes information such as the location, date, duration, shape, and description of the UFO that was spotted. Unfortunately, as is to be expected with so much data, the dataset first had to be cleaned as there were numerous missing values, mistakes, and other things that needed to be tweaked.

2.1 Exploratory Data Analysis

Our initial step in the data cleaning process was to check for any missing values in the dataset and remove the observations with missing values. We then reviewed the data to ensure the consistency and accuracy of variables such as state and city names, date and time of the sighting, and the duration of the sighting. To correct any inaccuracies in the longitude and latitude coordinates, we used additional resources to identify the correct city and state names. To gain insights into the trend of UFO sightings over time, we cleaned the data based on the total number of sightings per year, month, and hour in a day. This helped us determine whether the frequency of sightings was increasing over time or not. We also referenced outside resources regarding the military distribution, this helped us determine whether the frequency of sightings was related to distance to the military bases.

2.2 Methods

In order to address our hypotheses, we utilized graphical methods to aid in our analysis. For geographical correlation, a heatmap was selected as a suitable tool for investigating the potential correlation between UFO sightings and military bases. The heatmap depicts the frequency of UFO sightings across various regions in North America, while also highlighting the locations of military bases in red.

Interaction plots were found to be most useful for examining the differences among seasons. To ascertain the effects of the four seasons on the frequency of UFO sightings, we plotted the proportion of sightings accounted for by each season for every year from 1969-2019. We plotted the proportion accounted for largely due to the fact that the number of sightings from 1969 to around 2005 remained quite low. If we plotted sightings for all our data, the graph would have been scaled to the maximum number of sightings which was somewhere around 160 times the size of the minimum (7022 and 44, respectively); This would have made it nearly visually impossible to identify which season was most impactful up until 2005.

A relatively simple plot of the number of sightings vs the year was created to identify the correlation between those two variables.

2.3 First Hypothesis

How does geographical location correlate with the frequency of UFO sightings across the US?

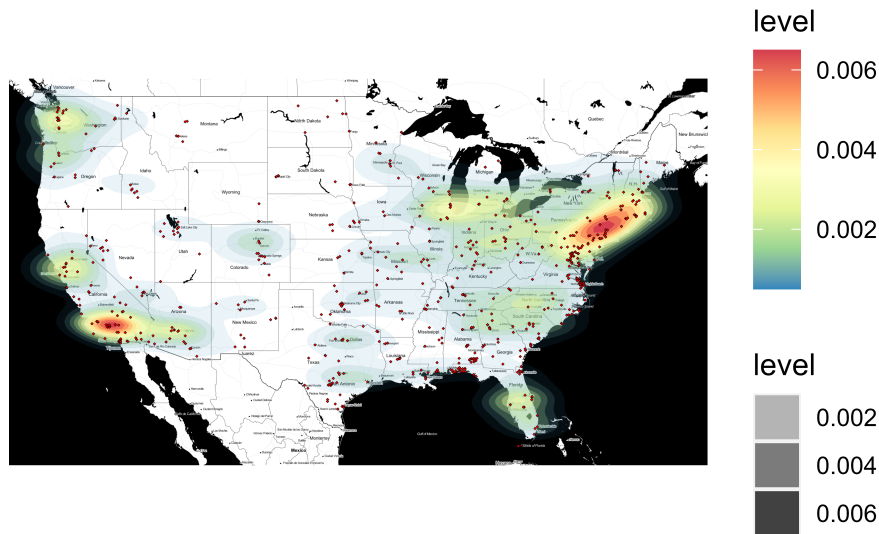


Figure 1: Heatmap of Military Bases Marked in Red

- We found that only a mere 4.3% of observations came from Hawaii, Alaska and Canada so we decided to restrict our analysis to sightings made in mainland US.
- The heatmap indicated that sightings were more frequent in areas with a higher concentration of military bases.
- There was a clear correlation between population density and the frequency of sightings, with larger cities exhibiting a higher number of sightings.

Overall, the analysis suggests that there may be certain factors that influence the frequency and distribution of UFO sightings in the United States. The positive correlation between military bases and sightings, as well as the association between population density and sightings, provides evidence for the idea that human activity and awareness may play a role in the occurrence of these events. Further research may be needed to fully understand the relationship between these factors and UFO sightings, and to explore other potential explanations for the observed patterns.

2.4 Second Hypothesis

How has the number of UFO sightings in the US changed over time? Why?

- There is a sudden spike in sightings in the year 2006, which is notably higher than any other year in the dataset.
- There is an increase in the variation of UFO sightings post-2006. This suggests that there was a greater degree of unpredictability in the occurrence of UFO sightings in the years following 2006.

These findings suggest that there may have been a significant event or events in 2006 that contributed to the sudden increase in UFO sightings, and that the frequency and distribution of sightings may have become more erratic in the years that followed. Further investigation may be needed to fully understand the reasons for these patterns, and to explore potential explanations for the sudden increase and subsequent variability in UFO sightings.

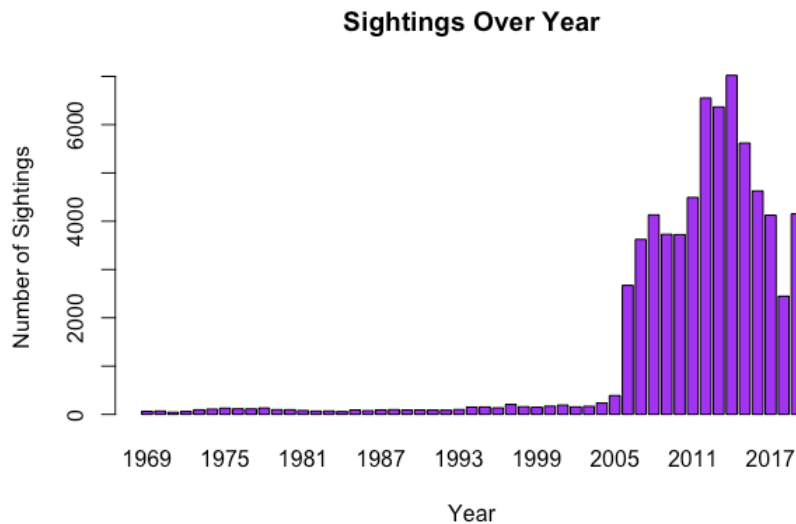


Figure 2: UFO Sightings Over time

2.5 Third Hypothesis

How do the four seasons differ in the proportion of UFO sightings they account for?

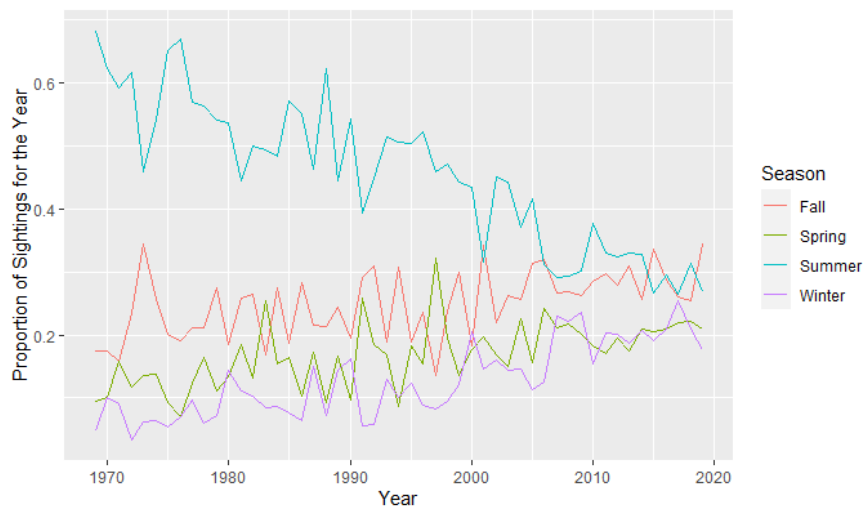


Figure 3: UFO Sightings Over Season

- The plot of UFO sightings over season showed that the frequency of sightings varied significantly depending on the season, with summer exhibiting a notably higher number of sightings compared to other seasons.
- The analysis also found that the effects of season on UFO sightings changed over time. While summer was initially the dominant season for sightings, this effect regressed to the mean over time, with the differences in frequency between seasons becoming less pronounced.

This suggests that there may be some underlying factors that contribute to the variation in UFO sightings across different seasons, but that these effects may change over time. These findings

highlight the complex and dynamic nature of UFO sightings, and suggest that there may be multiple factors at play that contribute to their occurrence and distribution.

3 Results

3.1 First Hypothesis

The analysis found a correlation between the frequency of UFO sightings and geographical location, with a higher concentration of sightings observed in areas with a greater number of military bases. This suggests that human activity and awareness may play a role in the occurrence of these events. However, the analysis was limited to sightings in mainland US, and it is unclear whether the same correlation would hold true for other areas. Further research may be necessary to fully understand the relationship between geographical location and UFO sightings.

3.2 Second Hypothesis

The analysis found a sudden spike in UFO sightings in 2006, followed by an increase in variability in the years that followed. It is unclear what caused this sudden increase in sightings, and further research may be needed to explore potential explanations. However, the findings suggest that the frequency and distribution of UFO sightings may be influenced by a variety of factors that can change over time, highlighting the complexity of this phenomenon.

3.3 Third Hypothesis

The analysis found that there were differences in the frequency of UFO sightings across the four seasons, with summer exhibiting a higher number of sightings compared to other seasons. However, the effects of season on UFO sightings changed over time, with the differences between seasons becoming less pronounced. This suggests that while season may play a role in the occurrence of UFO sightings, the influence of this factor may change over time. Further investigation may be needed to fully understand the reasons for these patterns.

4 Discussion

Upon having concluded our analysis of the data we were presented with, we found that certain areas of the country were far more populated with UFO sightings than others. Such areas consist of the West Coast and the East Coast. The impact of military bases appears a bit unclear. While many clusters of military bases do indeed exhibit some degree of heat on our map, there are many bases in the middle of the country which appear to be in locations which also have no UFO sightings. Moreover, the northwestern sector of Florida appears to be extremely populated with military bases, yet there seem to be little to no UFO sightings in this area. While we would like to say that military bases do demonstrate some sort of correlation with the frequency of UFO sightings in a given area, we cannot say this with conviction and thus our data may yet be inconclusive. More statistical testing and analysis would be necessary to come to a complete and accurate conclusion. Perhaps a linear/classification model handling the factors of local population, type of military base, proximity of sighting to military base, and would give a more accurate indication of how impactful military bases truly are on the number of UFO sightings reported each year.

With regards to how the seasons appear to affect the frequency of sightings throughout the year, this data interacted with time. Essentially, from the years 1969 through 2005, Summer accounted

for a significantly larger proportion of UFO sightings than all other seasons. However, once we hit around 2006 or 2007, this gap diminishes to the point of there being little difference amongst the four seasons with no single season accounting for more 38% of the reported sightings. Meanwhile, the smallest proportion accounted for was 16%. For comparison the minimum and maximum proportions accounted for from 1969 to 2005 were 5% and 68%, respectively. To determine the significance of these findings, a Tukey test should be conducted. By conducting a Tukey test, we would be able to further validate our findings regarding the impact of seasons on the frequency of sightings. This test would provide a more detailed understanding of the significance of the trends we have observed and could help to further inform our analysis and conclusions about the dataset.

As mentioned earlier in the paragraph, there was a noticeable increase in the number of UFO sightings around 2006 or 2007. However, the cause of this surge remains uncertain. One possible explanation is the rapid technological advancements during this time, with the release of the first iPhone in 2007. Although this is a speculative hypothesis, as we have not conducted sufficient research into other significant events that may have contributed to this increase, it is a plausible explanation.

Interestingly, after this spike in sightings, there appears to be a significant amount of variation in the number of reported sightings from year to year. Despite our fascination with this topic, we have yet to develop any theories to explain this variation. The reasons for this inconsistency feel inherently open-ended, and further research is necessary. It may be useful to investigate the impact of technological advancements and other potential factors that could have contributed to the post-2007 variation. Statistical tests could also be run to determine if there are significant differences in the number of sightings reported post-2007.

5 Conclusion

In conclusion, the analysis of UFO sightings in the US revealed a complex and dynamic phenomenon that is influenced by multiple factors, including geographical location, time, and season. The findings suggest that the frequency and distribution of UFO sightings may be influenced by human activity and awareness, as well as other factors that are not yet fully understood. The sudden spike in 2006 and the subsequent increase in variability highlight the need for further research to understand the reasons for these patterns. The correlation between the frequency of sightings and the number of military bases in an area suggests human activity and awareness may play a role. While season may affect sightings, the effects change over time, indicating that there may be other underlying factors yet to be understood. The study underscores the need for continued research into UFO sightings to better comprehend the nature of this phenomenon.

6 Reference

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